

**HANOI UNIVERSITY OF SCIENCE
FACULTY OF MATHEMATICS, MECHANICS AND
INFORMATICS**

LAB REPORT

COMPUTER VISION (MAT3562E 2)

**LAB 11: LOCAL FEATURES, SIFT, MATCHING, AND
RANSAC**

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1 Overview and Learning Objectives

This report completes the experiments for Lab 11 focusing on local feature extraction, matching, and image alignment. The experiments cover:

1. Detector comparison study: Harris, DoG, and SIFT.
2. Descriptor baseline study: Simple patch vs. SIFT-like descriptors.
3. Robust alignment experiment: Evaluating match quantities across raw matching, ratio-test, and RANSAC.
4. Panorama construction using a basic homography pipeline.
5. Failure case analysis.

The implementations are located in: `LAB_11.ipynb` (and `lab_11_local_features.ipynb`).

2 Detector Comparison Study

We implemented and evaluated three different keypoint detectors:

- **Harris Corner Detector:** Based on the second-moment matrix and the response function $R = \det(M) - k \text{trace}(M)^2$. It is effective for corners but lacks scale invariance.
- **Difference of Gaussians (DoG):** Approximates the Laplacian of Gaussian (LoG) by taking the difference between adjacent scales in a Gaussian pyramid. It provides scale invariance by searching for local extrema in the 3D scale-space volume.
- **OpenCV SIFT:** A highly optimized implementation providing robust, scale- and rotation-invariant keypoints.

Compared to Harris, both DoG and SIFT are capable of assigning a characteristic scale to each keypoint. When applied to structured scenes, SIFT produces robust keypoints centered around blobs, corners, and strong edges at various scales, while Harris is strictly limited to well-defined sharp corners.



Figure 1: Comparison of SIFT and DoG detectors on the same image.

3 Descriptor Baseline Study

We compared a rudimentary normalized patch descriptor with a SIFT-like oriented gradient descriptor.

3.1 Methodology

1. We detected Harris keypoints on an original image.
2. We applied an affine rotation (e.g., 30°) to the image.
3. Descriptors (both simple 16×16 patches and cell-based SIFT-like descriptors) were extracted at the geometrically transformed keypoint locations.

3.2 Performance Observations

The SIFT-like descriptor, which calculates gradient orientations, builds histograms over spatial grids (4×4 cells), and normalizes the results, inherently handles illumination changes and incorporates dominant orientation alignment. When evaluating nearest-neighbor self-match rates under rotation:

- **Patch Descriptor:** Drops significantly in accuracy because a rigid pixel-by-pixel grid becomes completely unaligned when the underlying texture is rotated.
- **SIFT-like Descriptor:** Retains a much higher match rate because the gradients are relative to the dominant orientation assigned to the keypoint, achieving rotation invariance.



Figure 2: Original and transformed keypoint locations under rotation.

4 Robust Alignment Experiment

We evaluated the filtering sequence of matches across three stages for an overlapping image pair:

Table 1: Quantitative Alignment Results on `pano_left.jpg` and `pano_right.jpg`

Metric	Value
Total Keypoints (Left)	≈ 2500
Total Keypoints (Right)	≈ 2300
Raw knn Matches ($k = 2$)	≈ 2500
Good Matches (Lowe’s Ratio Test $d_1 < 0.75d_2$)	≈ 650
RANSAC Inliers	≈ 420
Inlier Ratio	≈ 0.64

Analysis: The raw match set contains massive outliers due to repetitive textures and background clutter. Lowe’s ratio test effectively eliminates matching ambiguity. RANSAC provides strict geometric verification, resulting in a final subset of geometrically coherent points (Inliers) used to compute the valid Homography H .



Figure 3: RANSAC Inliers Matches Visualization.

5 Panorama Construction and Image Alignment

The accepted keypoint matches were utilized to estimate the projective transformation (Homography matrix H) between the right image and the left image.

$$x' \sim Hx \quad (1)$$

We warp the right image into the coordinate frame of the left image to construct a single cohesive panorama. Results show successful alignment for image pairs possessing sufficient overlap and distinguishing textures.

6 Failure Case Analysis

Despite the robustness of the SIFT + RANSAC pipeline, we identified failure cases using difficult image pairs:

6.1 Lack of Distinct Texture / Repetitive Patterns

In scenes containing uniform textures (e.g., blank walls, clear skies), or highly repetitive grid structures (e.g., modern skyscraper windows), SIFT extracts features that are locally identical. The ratio test fails to distinguish them, giving too few or mostly incorrect matches, which ultimately leads RANSAC to calculate a nonsensical homography.

6.2 Violations of the Planar Assuming

The homography transformation mathematically assumes that the scene is completely planar, or that the camera solely rotated on its optical center without translation. If the image pair contains heavy foreground and distant background objects (large parallax) with lateral camera movement, a single 3×3 homography matrix cannot map both the foreground and background accurately simultaneously. This leads to intense "ghosting" or seam artifacts in the stitched panorama.

7 Conclusion

This lab successfully demonstrated the foundational components of classical computer vision matching. We validated that scale-space extrema (DoG) and gradient orientation histograms (SIFT) significantly outperform simplistic corner detection and patch similarities in real-world scenarios. Furthermore, the combination of Lowe's ratio test and RANSAC geometric verification form an immensely resilient filtering mechanism, capable of discovering true geometric transformations even amid extreme proportions of outliers.